YAO MA

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EDUCATION

PhD in Quantum Information Technology, Computer Science Nov. 2019 - Dec. 2023
LIP6, Sorbonne Université, Paris, France

Engineer (Master), Engineering of Electronics and Digital Technology Sep. 2015 - Oct. 2017
École Polytechnique de l'Université de Nantes, Nantes, France
South China University of Technology, Canton, China (joint master degree)

Bachelor, Information Display and Optoelectronics Technology Sep. 2011 - Jul. 2015
South China University of Technology, Canton, China

PROFESSIONAL EXPERIENCE

Research Fellow Quantum Researcher April. 2026 - Now
Supervisors: Prof. Charles Lim, Department of Electrical and Computer Engineering, National University of Singapore, Singapore

- Lead the project "Relativistic Zero-Knowledge Proofs Systems for Space and Infrastructure". Develop system through both theoretical analysis and experimental implementation.

Senior Associate Quantum Researcher Mar. 2024 - Jan. 2026
Supervisors: Prof. Charles Lim, and Dr. Kaushik Chakraborty, Global Technology Applied Research, JPMorgan-Chase, Singapore

- Lead the project "Relativistic Zero-Knowledge Proofs for Identity Verification". Optimize system performance through both theoretical analysis and experimental implementation.
- Lead the project "Quantum-safe Distributed Ledger for Blockchain Application". Conduct formal security analysis and performance benchmarking to evaluate system robustness and efficiency.

PhD Thesis: Quantum Hardware Security and Near-term Applications Nov. 2019 - Dec. 2023
Supervisors: Prof. Elham Kashefi, Dr. Myrto Arapinis and Dr. Marc Kaplan (from Industry), LIP6, Sorbonne Université, Paris, France

- Research Interests: Quantum Computation, Quantum Information, Quantum Cryptography, Hardware Security Primitives, Physical Unclonable Function, Copy Protection Functions
- Research Skills: Cryptanalysis, Security Modelling and Proofing, System-level Behavioral Specification, Simulation, and Implementation.

System-on-Chip (SoC) System Design Engineer Feb. 2018 - Oct. 2019
Supervisor: Dr. Marc Kaplan, Veriqloud, Paris, France

- Develop hardware and software systems (SoC FPGAs) for multiparty quantum communication algorithms/protocols with different underlying architectures.
- Co-design quantum communication algorithms/protocols as part of Quantum Internet Alliance (QIA).

INTERNSHIP

Embedded System Developer Mar. 2017 - Oct. 2017
ESI Group, Nantes, France

- Develop APIs for validating sensor data processing within the Advanced Driver Assistance Systems (ADAS) emulator.
- Co-design algorithms of vehicular automation in different driving scenarios.

Engineering Project Oct. 2016 - Feb. 2017
Ecole Polytechnique de l'Université de Nantes, Nantes, France

- Design Internet-of-Things (IoT) networks of multiple devices with different hardware architectures.
- Benchmarking of IoT networks with different communication protocols and topology.
- Develop a distributed data storage system for devices in the network.

SoC Embedded System Intern Researcher Jul. 2016 - Oct. 2016
OFFIS e.V. Institut für Informatik, Oldenburg, Germany

- Develop and optimise the flight controlling system of drone from floating point to fixed point.
- Implement the algorithm with Matlab Simulink and simulate it in CAMeL-View.
- Participate in a 2-week summer school of "System-Level Design".

PUBLICATIONS AND PATENTS

A Practical Post-Quantum Distributed Ledger Protocol for Financial Institutions 2026

- Authors: Wei-Zhu Yeoh, Naresh Goud Boddu, **Yao Ma**, Shaltiel Eloul, Giulio Golinelli, Yash Satsangi, Rob Otter, Kaushik Chakraborty
- DOI:10.48550/arXiv.2603.05005

Quantum-Safe Identity Verification using Relativistic Zero-Knowledge Proof Systems 2025

- Authors: **Yao Ma**, Wen Yu Kon, Jefferson Chu, Kevin Han Yong Loh, Kaushik Chakraborty, Charles Lim
- DOI:10.48550/arXiv.2507.14324 (Under peer review at Nature Communications)

Quantum Lock: A Provable Quantum Communication Advantage 2023

- Authors: Kaushik Chakraborty, Mina Doosti, **Yao Ma**, Chirag Wadhwa, Myrto Arapinis, Elham Kashefi
- DOI:10.22331/q-2023-05-23-1014 (Publish in Quantum)

QEnclave - A practical solution for secure quantum cloud computing 2022

- Authors: **Yao Ma**, Elham Kashefi, Myrto Arapinis, Kaushik Chakraborty, Marc Kaplan
- DOI:10.1038/s41534-022-00612-5 (Publish in npj Quantum Information)

Patent**: Systems and methods for large modular multiplication using streaming interfaces** 2026

- Inventors: Kevin Loh, **Yao Ma**, Kaushik Chakraborty, Charles Lim, Marco Pistoia

Patent**: Method and server for delegated quantum computing using a hardware enclave** 2025

- Inventors: **Yao Ma**, Marc Kaplan

TALKS & POSTERS

QCrypt 2022 Aug. 2022

- Contributed Talk: Quantum Lock: A Provable Quantum Communication Advantage

Workshop: Quantum Information and the Frontiers of Quantum Theory Jun. 2022

- Presentation: Quantum Lock: A Provable Quantum Communication Advantage

The International Conference on Quantum Communication (ICQOM) Oct. 2021

- Contributed Talk: QEnclave - A practical solution for secure quantum cloud computing

QCrypt 2021 Aug. 2021

- Poster: QEnclave - A practical solution for secure quantum cloud computing

SKILLS

Language	Chinese and Cantonese (Native), English (Professional), French (Intermediate)
Software	Vivado, VS Code, Matlab, Mathematica
Programming Language	Verilog HDL, C, C++, Python, Wolfram language
Interests	Robotics, Tennis, Hiking, Boardgames